ASMI BASNET

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PROFESSIONAL SUMMARY

Applied Data Scientist and Graduate Researcher (Saint Louis University M.S. in Analytics candidate; B.E. in Electronics & Communication) with over three years of experience building statistical models, automated analytical pipelines, and document-intelligence systems. I specialize in carrying complex analytical questions from raw, messy data into defensible, reproducible answers that withstand scrutiny. Experienced working across Python, SQL, and cloud environments (AWS S3, Aurora/PostgreSQL architectures), with a strong background hardening exploratory notebooks into reviewed, tested pipelines. Proven track record benchmarking supervised models (XGBoost, LightGBM, Random Forest) against rigorous baselines, developing time-series anomaly detection engines, architecting auditable NLP/RAG document verification suites (PolicyGuard), and communicating findings through clear visual narratives for institutional decision-makers. Eager to support mission-critical federal analytics at Peregrine Advisors.

TECHNICAL SKILLS & CORE COMPETENCIES

Statistical Modeling & ML: Supervised & Unsupervised Machine Learning, Statistical Hypothesis Testing, Time-Series Forecasting (Prophet, LSTM, ARIMA), Classification, Anomaly Detection, Model Validation (K-Fold Cross-Validation, ROC-AUC, PR-AUC, F1), Error Analysis, Baseline Benchmarking, Feature Engineering.
Programming & Query Languages: Python (Pandas, NumPy, Scikit-Learn, PyTorch, SQLAlchemy, FastAPI), SQL (Complex CTEs, Analytical Window Functions, Multi-Table Joins, Query Optimization), PostgreSQL, DuckDB, Snowflake, SQLite, C/C++, R, Statistical Analysis (Regression, ANOVA, Chi-Square).
Cloud & Data Engineering (AWS): Amazon Web Services (AWS) Data Platforms Familiarity (Amazon S3, Amazon Aurora / Relational Databases, Cloud Pipelines), Containerization (Docker, Docker Compose), Linux/Bash Scripting, Git / GitHub Version Control, Automated Testing & Verification Loops.
AI, Document Intelligence & Governance: Foundation Models & LLM Integration (Amazon Bedrock / Claude & OpenAI APIs), Document Parsing, Offline RAG Suites, Citation Verification, Independent Verification & Validation (IV&V), Data Governance, Power BI, Tableau, Excel (Modeling, VBA), Technical Writing.

PROFESSIONAL EXPERIENCE

Saint Louis University — **St. Louis, MO**
Graduate Researcher, SPS Corps – Analytics Initiative — *August 2025 – Present*

- Frame core analytical questions with departmental leaders, turning high-level operational inquiries into testable statistical designs and automated data pipelines across 8+ operational databases.
- Refactored complex SQL queries by replacing nested subqueries with modular common table expressions and window functions (ROW_NUMBER, DENSE_RANK, LAG/LEAD), reducing query runtimes by 40% and standardizing data definitions.
- Developed a hybrid forecasting engine combining Facebook Prophet with PyTorch LSTM neural networks, integrating automated anomaly detection (flagging deviations beyond $\pm 2.5\sigma$) to deliver defensible quarterly budget and workload projections.
- Hardened exploratory analysis scripts into reviewed, reproducible pipelines and implemented automated data quality checks (IV&V) that enforce referential integrity and catch schema drift before records reach reporting layers.
- Designed interactive Power BI dashboards and executive briefings, presenting complex statistical trends and model limitations in accessible plain language to university leadership.
- Authored rigorous technical runbooks, data dictionaries, and model evaluation reports, conducting hands-on training sessions for over 25 administrative staff members to ensure organizational continuity.

Saint Louis University — **St. Louis, MO**
Data Analyst Intern / BI Developer — *June 2024 – August 2024*

- Built and deployed 6 interactive Power BI and Tableau dashboards backed by clean dimensional star schemas, consolidating 15+ institutional KPIs to give administrators clear visibility into operational performance.
- Engineered automated data profiling, cleaning, and reconciliation scripts in Python (Pandas) and SQL across 10,000+ transaction records, setting up automated webhook alerts to immediately flag data discrepancies.
- Enforced strict institutional data governance standards, establishing row-level security (RLS) and audit logging routines to ensure sensitive academic records remained fully compliant with privacy regulations.

Entegra Sources Pvt. Ltd. — **Kathmandu, Nepal (Remote)**
Data Analyst / Researcher & Systems Analyst — *December 2022 – July 2023*

- Extracted, cleansed, and normalized multi-source operational and survey datasets (5,000+ records) into standardized relational schemas using Python (Pandas, SQLAlchemy) and SQL, guaranteeing complete data consistency.
- Engineered automated validation scripts and exception-handling routines in Python and C/C++, implementing boundary tests and audit checks to maintain a 99.8% pipeline reliability rate across recurring analysis runs.
- Authored formal methodology documentation, data dictionaries, and statistical audit reports (SPSS/R regression, ANOVA, chi-square) to ensure transparent, peer-reviewable research standards.

FEATURED APPLIED MACHINE LEARNING & DOCUMENT INTELLIGENCE PROJECTS

PolicyGuard — Document Intelligence & Automated Citation Verification Suite — **Python, Ollama, ChromaDB, FastAPI, Docker, Streamlit**

- Engineered a document-intelligence compliance application designed to analyze complex regulatory and policy corpora locally with zero external API exposure, keeping sensitive agency documents strictly protected on premises.
- Implemented automated chain-of-thought citation verification that evaluates retrieved context sentence-by-sentence, flagging contradictory requirements (FLAG_CONFLICT) and missing evidence (CLARIFY) to produce an auditable governance health score.

WashU Clinical Analytics & Predictive Risk Modeling Datathon — **Python, Scikit-Learn, LightGBM, XGBoost, Hyperopt**

- Built an end-to-end predictive modeling pipeline across 62,000+ records, engineering 59 domain features (visit frequencies, variance metrics, interaction terms) to identify elevated-risk patient cohorts.
- Benchmarked Logistic Regression, Random Forest, LightGBM, and XGBoost against baseline models using stratified K-fold cross-validation and hyperparameter optimization, achieving an AUC of 0.8520 with documented error distributions.

SCD Type-2 Dimensional Data Warehouse & Reproducible Analytics Engine — **Python, SQL, DuckDB, Star Schema, Kimball Modeling**

- Designed a Kimball dimensional data warehouse implementing Slowly Changing Dimensions (SCD Type-2) across 10,000+ records with surrogate keys and date tracking, guaranteeing a complete point-in-time audit trail for defensible retrospective analysis.

EDUCATION & PROFESSIONAL CREDENTIALS

Saint Louis University | Master of Science in Analytics (STEM) — **St. Louis, MO | Expected May 2026**
Tribhuvan University | Bachelor of Engineering in Electronics & Communication (STEM) — **Kathmandu, Nepal | Conferred 2023**
Certifications: Project Management & Agile Certification (GlobalShala & SLU) · AI Engineer for Data Scientists Associate (DataCamp) · Building with Claude & Anthropic API (Prompt Caching & Tools).